12-20 第1篇

阅读理解（说明文） Why Are Winter Sunsets So Orange?

① On a clear winter afternoon, somewhere between **16:40 and 17:20** in northern China, you may step to a west-facing window and notice, almost with small surprise, that the sun hanging low over the rooftops is not its usual noon **white-yellow**. It is **deep orange**, sometimes even a quiet **red-pink**. The sky around it glows in soft ribbons of peach, salmon, and violet. Children on the pavement below stop for a moment to look up. Adults quietly take out their phones. The whole world, for **nine or ten minutes**, seems to be painted by a careful hand. The strange thing is: the sun itself is exactly the same temperature and brightness it was at noon. The sky is the same sky. Only the **angle** has changed. This short, beautiful winter show is one of the clearest demonstrations of a small physics rule called **atmospheric scattering** — first measured carefully by the British scientist **Lord Rayleigh in 1871**.

② **Finding one: white sunlight is secretly a mixture.** Sunlight looks white to our eyes, but when it passes through a glass prism (or a raindrop), it separates into the familiar seven-colour band of the rainbow: red, orange, yellow, green, blue, indigo, violet. Each colour travels as a light wave with a different **wavelength** (the distance between two wave crests). Red light has the **longest** wavelength — about **700 nanometres**. Violet light has the **shortest** — about **400 nanometres**. Blue light is close to the violet end, at about **450 nanometres**. This tiny size difference, no bigger than the width of a few molecules, turns out to matter enormously when sunlight meets the thin clear gas we call **air**.

③ **Finding two: short blue waves scatter more than long red waves.** Earth's atmosphere is about **99%** nitrogen and oxygen molecules, each one roughly **0.3 nanometres** across. When a ray of sunlight enters the atmosphere, the tiny gas molecules act like extremely small invisible bumpers. A simple physics calculation (done first by Rayleigh) shows that the **amount of scattering goes up very sharply** as wavelength goes down — precisely as **1 divided by the fourth power of the wavelength**. In plain numbers, **violet and blue light scatter roughly 10 times more strongly than red light**. So the short blue-violet part of sunlight gets bounced in every direction, while the longer red-orange part continues almost straight. This is **exactly why the daytime sky is blue**: the blue that your eye sees is scattered sunlight reaching you from every empty bit of sky, not from the sun itself.

④ **Finding three: at sunset, sunlight travels through much more air.** At noon, when the sun is directly overhead, sunlight passes through about **100 km** of effective atmosphere before reaching you. But at sunset, when the sun is nearly on the horizon, the light must travel **almost sideways through the thick lower atmosphere**, a path of around **1,200 km** — **twelve times longer**. On this long sideways journey, something quietly dramatic

happens: **all the blue and green light gets scattered out of the ray** long before it reaches your eye. What remains is mostly the **yellow, orange, and red** — the long-wavelength survivors that did not scatter. By the time the sunlight lands on the west-facing window, it has been, in a sense, **filtered** by 1,200 km of air into a warm late-afternoon orange.

⑤ **Why is winter orange especially rich?** Because cold, dry winter air is **unusually clean**. Warm, humid summer air is full of tiny water droplets, pollen, and small organic dust — all of which scatter **all colours fairly evenly**, making summer sunsets a duller milky-orange. Cold winter air, by contrast, has very few of these large particles. The long sunset path is then dominated by **only Rayleigh scattering** — which is sharply favourable to reds and oranges. A 2021 Peking University atmospheric science study of **3,204 winter and summer days** in eight Chinese cities measured afternoon sky colour and found a **17% richer red-orange intensity** in clean winter sunsets than in humid summer sunsets of matched brightness. Winter, quietly, is the **clean season for sunlight**.

⑥ **What about the pink-salmon ribbons in the sky above the sun?** Those are also explained by scattering, with one extra twist. The sunset light that passes just **above** the sun — not straight at your eye, but at a small angle — scatters one more time off the dust and ice crystals high in the **upper atmosphere**, redirecting part of its warm colours down toward you. The higher, thinner ice clouds (**cirrus** clouds, made of tiny ice needles at 8—12 km altitude) act as a high kind mirror, turning part of the orange-red into softer pink. That is why the **prettiest** winter sunsets are often the ones with a thin wispy cloud layer; a completely empty sky gives only the red-orange of the sun itself, which is beautiful but simpler.

⑦ **What about the strange green flash?** On very clean, dry, flat horizons (rare over cities, more common at sea), in the last half-second before the sun disappears, an observer may see a **tiny green flash** at the very top edge of the sun. This is not a myth. It is the result of the atmosphere acting as a weak prism: the last edge of the sun bends its colours slightly, and because blue has already been scattered away, the next-shortest survivor — green — briefly peeks out on top. Jules Verne wrote a whole 1882 novel about it. A 2012 photographic study on the Chilean coast caught the green flash in **2.3%** of clean sunsets.

⑧ **Why don't winter sunrises look the same as winter sunsets?** They actually do — in physics. The same long atmospheric path and the same Rayleigh scattering happen at sunrise. But sunrise often looks **slightly paler orange** for a small human reason: in the early morning the air is often full of **fine cold water-vapour fog** that hasn't yet burned off, and this fog scatters colours a bit more evenly than clean afternoon air. That is why poets often find **winter sunsets more vivid** — not because sunset sunlight is different, but because afternoon winter air is usually cleaner than morning winter air.

⑨ **The quiet lesson.** The next time the afternoon sky in late December turns, for about ten careful minutes, that astonishing peach-orange-pink, remember: you are not

watching magic. You are watching, with your own eyes, **a 1,200 km journey of sunlight through a clear cold atmosphere, in which blue and violet have been quietly bounced away and only red, orange, and yellow have survived the trip.** The physics is older than the human species; the sight is new every evening. And for ten kind minutes, as the sun dips into its horizon of bare winter trees, you are the person for whom 150 million km of travelling light made its final, softest, slowest bend.

注释： atmospheric scattering 大气散射；wavelength 波长；nanometre 纳米；prism 棱镜；cirrus 卷云；humid 潮湿的；green flash 绿色闪光

根据短文内容，选择最佳答案。

1. According to Paragraph ②, what is the key relationship between colour and wavelength of light?

A. All colours of light have the same wavelength.
B. Red light has the shortest wavelength and violet the longest.
C. White sunlight is secretly a mixture that separates through a prism or a raindrop into the seven-colour rainbow band (red, orange, yellow, green, blue, indigo, violet); each colour travels as a light wave with a different wavelength, with red being the longest at about 700 nanometres, blue near the violet end at about 450 nanometres, and violet the shortest at about 400 nanometres — this tiny size difference matters enormously when sunlight meets air.
D. Only red light exists in sunlight.

2. How does Paragraph ③ explain why the daytime sky is blue?

A. The sky is blue because the ocean reflects blue light upward.
B. The sky is blue because the sun is actually blue at noon.
C. Earth's atmosphere is about 99% nitrogen and oxygen molecules, each roughly 0.3 nanometres across; Lord Rayleigh's calculation (1871) shows that scattering amount rises as 1 divided by the fourth power of wavelength, meaning violet and blue light scatter roughly 10 times more strongly than red light, so the short blue-violet part of sunlight gets bounced in every direction while the longer red-orange part continues almost straight — the blue the eye sees is therefore scattered sunlight reaching you from every empty bit of sky, not from the sun itself.
D. The sky is blue because humans evolved to see only blue.

3. What is the physical difference between sunlight at noon and sunlight at sunset, and why does this make the sunset orange (Paragraph ④)?

A. Sunset light is hotter than noon light and therefore turns orange.
B. Sunset sunlight is yellow because the sun changes colour in the evening.
C. At noon, sunlight passes through about 100 km of effective atmosphere, but at sunset

the sun is nearly on the horizon and the light must travel almost sideways through about 1,200 km — twelve times longer; on this long journey, all the blue and green light gets scattered out of the ray long before it reaches your eye, and what remains is mostly the yellow, orange and red — the long-wavelength survivors — so the sunlight arrives filtered by 1,200 km of air into a warm late-afternoon orange.
D. Sunset light contains extra red light newly made at the horizon.

4. Why are winter sunsets especially rich, and what do pink-salmon "ribbons" above the sun come from (Paragraphs ⑤ and ⑥)?

A. Winter sunsets are duller than summer sunsets because of snow.
B. Cold dry winter air is unusually clean — with very few of the water droplets, pollen and organic dust of humid summer air — so the long sunset path is dominated by only Rayleigh scattering, which sharply favours reds and oranges; a 2021 Peking University study of 3,204 days in eight Chinese cities measured a 17% richer red-orange intensity in clean winter sunsets than in humid summer sunsets of matched brightness; and the pink-salmon ribbons above the sun come from sunset light that passes just above the sun at a small angle and scatters one more time off high-altitude dust and ice crystals, especially thin cirrus ice clouds at 8—12 km altitude, which act as a high kind mirror redirecting some warm colours downward as softer pink — so the prettiest winter sunsets often have a thin wispy cloud layer.
C. All cold air scatters colours more evenly than warm air.
D. Pink ribbons come from a layer of pollution at the horizon.

5. What is the main idea of the final paragraph?

A. Late December sunsets are magic and unexplainable.
B. When the late-December afternoon sky turns, for about ten careful minutes, an astonishing peach-orange-pink, the viewer is watching not magic but a 1,200 km journey of sunlight through a clear cold atmosphere — blue and violet quietly bounced away, only red, orange, and yellow surviving the trip; the physics is older than the human species, the sight is new every evening, and for ten kind minutes, as the sun dips into its horizon of bare winter trees, the viewer is the person for whom 150 million km of travelling light made its final, softest, slowest bend.
C. Winter sunsets should be avoided because they hurt the eyes.
D. Only trained scientists may enjoy sunsets.

答案与解析

1.
【答案】C
【题型】细节理解题
【定位】第②段白光 + 波长
【解析】"红 700 / 紫 400 / 蓝 450 nm"直引。A、B、D 均错。故选 C。

2.
【答案】C
【题型】细节理解题
【定位】第③段瑞利散射
【解析】"99% 氮氧 / $1/\lambda^4$ / 蓝紫散射 10 倍 / 天蓝来自四方散射"直引。A、B、D 均错。故选 C。

3.
【答案】C
【题型】细节理解题
【定位】第④段黄昏光程
【解析】"正午 100 km / 黄昏 1200 km / 蓝绿已散 / 红橙黄幸存"直引。A、B、D 均错。故选 C。

4.
【答案】B
【题型】细节理解题
【定位】第⑤⑥段冬天干净 + 卷云
【解析】"冬空气洁净 / 17% 更浓 / 卷云 8–12 km 反射出粉色"直引。A、C、D 均错。故选 B。

5.
【答案】B
【题型】主旨大意题
【定位】第⑨段结尾
【解析】"1200 km 光旅 / 蓝紫被弹走 / 1.5 亿 km 光最柔最慢一弯"主旨。A、C、D 均偏离。故选 B。